Phillip J. Macias

Computational Systems Modeler, Mountain View, CA

Email | LinkedIn | Website | GitHub

Summary

Computational systems modeler with 10+ years of experience designing and executing large-scale simulation frameworks in collaborative, deadline-driven environments. Experienced in distributed HPC workflows, high-dimensional parameter exploration, and uncertainty quantification and propagation. Contributed to simulation infrastructure for a NASA flagship mission.

Technical Expertise

- **Programming:** Python, C, Fortran, Bash | **HPC:** Slurm, distributed simulation campaigns, large-scale parameter sweeps
- **Modeling:** Monte Carlo methods, uncertainty quantification, sensitivity analysis, XGBoost, Explainable Boosting Machines (EBM), PyTorch (CNNs, Grad-CAM)
- **Data & Workflow:** Jupyter, NumPy, SciPy, Pandas, Matplotlib, SHAP, DuckDB, geospatial data (raster/HDF5), Git, reproducible pipelines

Experience

University of California, Santa Cruz *2012 – 2026*

Ph.D. Researcher (2012–2019); Postdoctoral Researcher (2020–2026)

Computational Modeling & Simulation

- Contributed modeling deliverables within a **35-member** Science Investigation Team for the **NASA Nancy Grace Roman Space Telescope**; core member of a **6-person** simulation subgroup.
- Designed and executed distributed simulation campaigns spanning **thousands of model realizations** on Slurm-managed HPC clusters, producing technical reports used for NASA mission planning.
- Designed and ran large-scale **hydrodynamic simulations** using the FLASH AMR code; executed multi-day runs across **hundreds of CPU cores** to investigate nonlinear stability.

Projects

Flashpoint — Wildfire Severity Classifier *2026*

- Built an end-to-end ML pipeline predicting whether an active wildfire escalates or stays contained, using only the first 1–2 days of fire and weather data, from weather, topography, and vegetation features.
- Compared multiple modeling approaches (XGBoost, Explainable Boosting Machine, CNN with Grad-CAM); the **EBM** reached **82.6% accuracy (0.754 macro-F1)** on the binary escalation task, validated via leave-one-year-out cross-validation.

Mentorship & Leadership

- Mentored **15+** students (high school through graduate level) in computational modeling and research workflows, including a paid remote research cohort of **five** at The Thacher School.

Research Output

Co-author of **20** peer-reviewed publications in astrophysics and computational modeling.

Education

Ph.D., Astronomy & Astrophysics — University of California, Santa Cruz *2019*
M.S., Astronomy & Astrophysics — University of California, Santa Cruz *2015*
B.S., Physics (Highest Honors) — University of California, Santa Barbara *2012*